

Draw It or Lose It

CS 230 Project Software Design Template

Version 1.0

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Document Revision History

Version	Date	Author	Comments
1.1	10/17/26	K	Updated recommendations for Project Three, including storage, memory, distributed systems, and security.

Executive Summary

The Gaming Room currently offers its game, Draw It or Lose It, as an Android application and would like to expand it into a web-based game that can be accessed across multiple platforms. The new application will support multiple teams and players while ensuring that game and team names remain unique. To meet these needs, the application will use a web-based distributed design and software design patterns to efficiently manage games, teams, and players while allowing users on different platforms to access the game.

Requirements

The Gaming Room requires Draw It or Lose It to be expanded from its current Android application into a web-based game that can support multiple platforms. The application must allow each game to have one or more teams and each team to have multiple players. Game and team names must remain unique so users can determine whether a name is already in use. The application must also ensure that only one instance of the game service exists in memory while using unique identifiers to manage games, teams, and players.

Design Constraints

The main design constraints for Draw It or Lose It include supporting multiple platforms, operating in a web-based distributed environment, maintaining unique game and team names, and managing multiple users at the same time. Since the application needs to work across different platforms, it should be designed so users can access the game through a web browser without requiring a separate version for each operating system. The application must also communicate over a network, which means the development team needs to consider reliable connections and communication between the client and

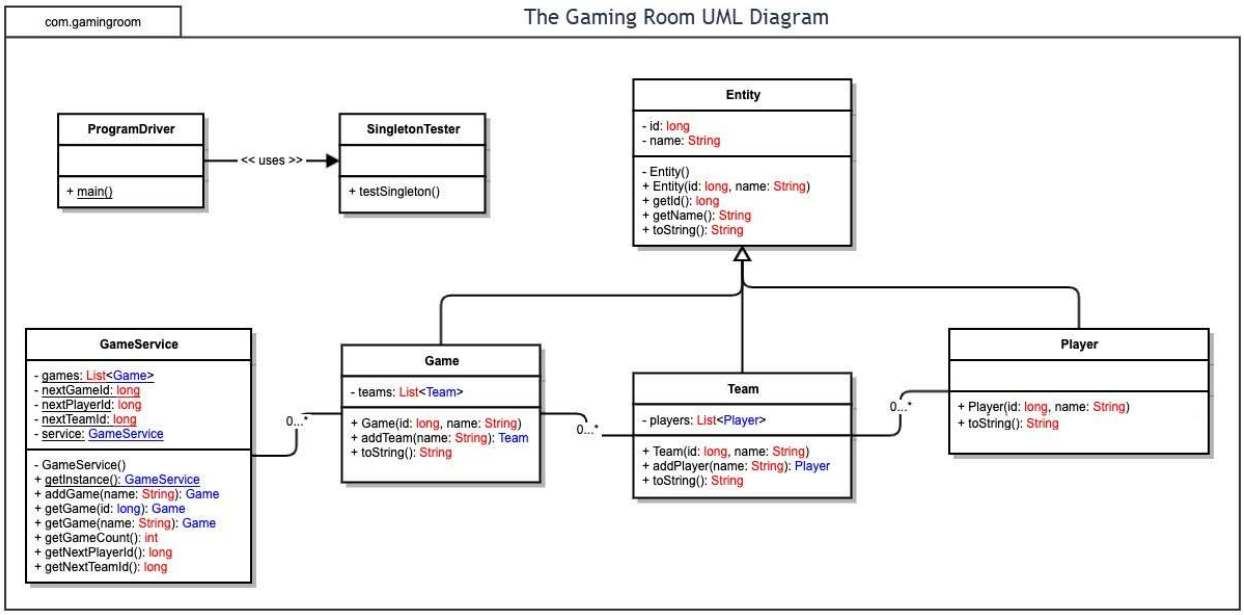
server. The design must prevent duplicate game and team names and properly manage games, teams, and players so that multiple users can interact with the application without interfering with one another.

System Architecture View

Please note: There is nothing required here for these projects, but this section serves as a reminder that describing the system and subsystem architecture present in the application, including physical components or tiers, may be required for other projects. A logical topology of the communication and storage aspects is also necessary to understand the overall architecture and should be provided.

Domain Model

The UML class diagram shows the relationships between the main classes used in the Draw It or Lose It application. The Entity class acts as a base class that stores the common id and name attributes. The Game, Team, and Player classes inherit these attributes from Entity, which demonstrates inheritance and prevents the same information from having to be repeated in each class. A Game can contain multiple teams, and each Team can contain multiple players. The GameService class manages the games and uses the Singleton pattern so that only one instance of the service exists in memory. The application also uses encapsulation by keeping class attributes private and providing methods to access or manage the data. These object-oriented principles help organize the application, reduce repeated code, and support the requirements for managing unique games, teams, and players.



Evaluation

Using your experience to evaluate the characteristics, advantages, and weaknesses of each operating platform (Linux, Mac, and Windows) as well as mobile devices, consider the requirements outlined below and articulate your findings for each. As you complete the table, keep in mind your client's requirements and look at the situation holistically, as it all has to work together.

In each cell, remove the bracketed prompt and write your own paragraph response covering the indicated information.

Development Requirements	**Mac**	**Linux**	**Windows**	**Mobile Devices**

Server Side	macOS can support a web-based application and can run web server software such as Apache or other third-party server tools. This would allow Draw It or Lose It to be hosted on a Mac and accessed by users through a web browser. However, Apple has discontinued macOS Server, making macOS less focused on dedicated server deployment than other operating systems. Using macOS would also require Apple hardware, which could increase the cost of expanding the server as the number of players grows. Although macOS could host the application, these limitations should be considered when planning a server that may eventually need to support thousands of users.	Linux is well suited for hosting web-based applications and offers server-focused distributions such as Ubuntu Server. It supports common web server software such as Apache and can be deployed on physical servers, virtual machines, or cloud environments. Linux is also designed to scale efficiently, making it a strong option for Draw It or Lose It as the number of players increases. Another advantage is cost because distributions such as Ubuntu can be used without operating system licensing fees, although paid support is available if needed. One weakness is that managing a Linux server may require developers or administrators who are comfortable working with Linux and command-line tools.	Windows is a strong option for hosting web-based applications because Windows Server provides tools designed specifically for server environments. It includes Internet Information Services (IIS), which can host websites, services, and applications and would support a web-based game such as Draw It or Lose It. Windows Server can also support larger deployments as the number of players increases. One advantage is its integration with Microsoft technologies and its familiar graphical management tools. However, Windows Server requires commercial licensing, which can increase costs compared with a free Linux server. The development team would need to consider these licensing costs as the server environment grows.	Mobile operating systems such as Android and iOS are primarily designed to run client applications rather than host large web-based applications. Although a mobile device can technically provide some network services, it would not be practical for hosting Draw It or Lose It, especially if the game needs to support thousands of players. Mobile devices have more limited processing power, memory, storage, and network reliability than dedicated servers. Instead, mobile devices should connect to a server hosted on another operating system and access the game through a responsive web browser. This allows the server to handle the main processing and data while the mobile device acts as the client.

Client Side	Mac users can access Draw It or Lose It through a web browser, so the development team would not need to create a completely separate application specifically for macOS. The responsive HTML interface should be tested with browsers available on Mac, especially Safari, to make sure that the game displays and functions correctly. Developers should also test different screen sizes and browser versions for compatibility. Using a web-based client can reduce development time and cost because much of the same client-side code can be shared across platforms. However, additional testing is still necessary to identify browser-specific problems and provide Mac users with a consistent experience.	Linux users can access Draw It or Lose It through a web browser, which means the development team would not need to build a separate Linux version of the game. The responsive HTML interface should be tested with commonly used browsers on Linux to make sure the game functions and displays correctly. Since Linux has many different distributions and configurations, developers may need additional testing to identify compatibility issues. However, sharing the same web-based client across platforms can reduce development time and cost. Developers with experience in web development and cross-browser testing would be important for providing Linux users with a consistent experience.	Windows users can access Draw It or Lose It through a web browser without requiring the development team to create a separate Windows application. The responsive HTML interface should be tested in commonly used browsers such as Microsoft Edge to ensure that the game displays and functions correctly. Since the same web-based client can be shared with other operating systems, this approach can reduce development time and cost. Developers will still need to perform cross-browser testing and account for different screen sizes, browser versions, and Windows configurations to provide users with a consistent experience.	Mobile devices can access Draw It or Lose It through a responsive web browser, allowing the same web-based application to support both iOS and Android devices. The development team will need to make sure the interface adapts to different screen sizes, orientations, and touch controls so the game remains easy to use on phones and tablets. Testing should be performed across different mobile devices and browsers to identify compatibility issues. A responsive web application can reduce development time and cost compared with creating separate native applications for iOS and Android, although additional mobile testing and expertise will still be necessary.

Development Tools	Mac development can use tools such as **Xcode** for Apple-specific development and cross-platform tools such as Java and common web-development IDEs. Apple provides Xcode through the Mac App Store, while distributing software through Apple's developer ecosystem can involve membership requirements and costs. For *Draw It or Lose It*, however, the browser-based design means the team can primarily use standard web technologies rather than maintaining a separate native Mac application.	Linux supports a wide variety of development tools and programming languages, including Java, JavaScript, Python, and common web-development frameworks. Many Linux development tools are open source and available without licensing costs, which can help reduce development expenses. Linux also provides command-line tools that are useful for configuring, testing, and deploying web applications. However, developers who are unfamiliar with Linux may require additional training or experience to work efficiently with the platform.	Windows supports development tools such as **Visual Studio**, Visual Studio Code, Java development environments, and other tools used for web development. Microsoft provides several editions of Visual Studio, including the free Community edition for qualifying users and organizations, while Professional and Enterprise editions require paid licensing. Because Windows supports many commonly used development tools, developers may already have experience working in its development environment.	Mobile development can require platform-specific tools such as **Xcode** for iOS and **Android Studio** for Android. Maintaining separate native applications could require developers with expertise in both platforms and increase development time and cost. Because *Draw It or Lose It* is being designed as a responsive web application, the team can instead use shared HTML, CSS, and JavaScript code for much of the client experience. Mobile browsers and developer tools can then be used to test the game on different devices, screen sizes, and operating systems.

Recommendations

Analyze the characteristics of and techniques specific to various systems architectures and make a recommendation to The Gaming Room. Specifically, address the following:

1. **Operating Platform**: Linux, specifically Ubuntu Server, would be an appropriate operating platform for hosting Draw It or Lose It. Linux is commonly used for server environments and can support the web server, application services, and database needed for the game. It can also be scaled as the number of players increases without requiring The Gaming Room to purchase Windows Server licenses. The server can host the main game services while users on Windows, macOS, Linux, iOS, and Android access the game through a responsive web browser. This allows The Gaming Room to support multiple computing environments without developing a completely separate version of the game for every operating system.

2. **Operating Systems Architectures**: Ubuntu Server supports several processor architectures, including 64-bit Intel and AMD systems and ARM64 systems. A 64-bit architecture would be appropriate for the Draw It or Lose It server because it can support the memory and processing requirements of a growing web application. Linux separates applications from the underlying hardware through the operating system and kernel, allowing the system to manage processes, memory, storage, networking, and hardware resources. Ubuntu Server can also operate on physical hardware, virtual machines, or cloud infrastructure, giving The Gaming Room flexibility as the game expands.

3. **Storage Management**: The Gaming Room should use persistent server storage with a Linux file system, such as ext4, along with a database for information that needs to be stored and retrieved by the application. Since Draw It or Lose It will have 200 high-definition images at approximately 8 MB each, the image library alone will require about 1,600 MB, or 1.6 GB, of storage. Additional storage will also be needed for application files, user information, games, teams, players, logs, and future updates. The file system can manage application files and other server resources, while the database can organize player and game information for efficient retrieval. The image library should be organized so individual images can be accessed quickly without searching through every file. Regular backups should also be maintained so information can be recovered if hardware fails or data becomes corrupted. As the application grows, additional or cloud-based storage could be added without requiring changes to the client devices.

4. **Memory Management**: Linux manages memory by allocating memory to active processes and using virtual memory so applications can operate without directly managing physical memory. Paging allows portions of programs and data to be moved between physical memory and secondary storage when necessary. For Draw It or Lose It, the operating system would allocate memory to the web server, game services, database, and other processes while keeping their memory spaces protected from one another. Since the application needs to display high-definition images at a steady rate, it should load only the images needed for the current game or round into RAM rather than loading the entire 1.6 GB image library at once. Frequently used images could also be cached, and upcoming images could be preloaded to reduce delays and help ensure drawings are fully rendered within the required 30 seconds. Once images are no

longer needed, their memory can be released to make room for other game sessions. As more players connect, the server should have enough physical memory to prevent excessive paging, which could reduce application performance.

5. **Distributed Systems and Networks**: Draw It or Lose It can operate as a distributed system by placing the main game logic and data on the Linux server while clients communicate with it across a network. Players using Windows, macOS, Linux, iOS, or Android could access the responsive web application through their browsers and exchange information with the server using standard web protocols and APIs. This approach allows the server to maintain shared game information while clients provide the user interface. Reliable network connectivity is important because an outage between a client and server could interrupt gameplay or prevent information from being updated. The system should therefore include error handling, monitoring, and recovery procedures so temporary connection problems or server failures do not unnecessarily damage or lose game data.

6. **Security**: Security should be included throughout the design of Draw It or Lose It rather than added after development. Users should be authenticated before accessing protected information, and authorization should limit users to only the resources and actions they need. This follows the principle of least privilege and reduces the amount of information that could be exposed if an account is compromised. Communication between clients and the server should use encrypted HTTPS connections to protect information while it travels across the network. On the Ubuntu server, firewalls can restrict unnecessary network traffic, while AppArmor can limit what individual applications are allowed to access. Sensitive stored information should also be encrypted when appropriate, and the server should be regularly updated, monitored, and backed up. These protections help maintain the confidentiality, integrity, and availability of user and game information.